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Multimodal deep learning model for prognostic prediction in cervical cancer receiving definitive radiotherapy: a multi-center study

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For patients with locally advanced cervical cancer (LACC), precise survival prediction models could guide personalized treatment. We developed and validated CerviPro, a deep learning-based multimodal prognostic model, to predict disease-free survival (DFS) in 1018 patients with LACC receiving definitive radiotherapy. The model integrates pre- and post-treatment CT imaging, handcrafted radiomic features, and clinical variables. CerviPro demonstrated robust predictive performance in the internal validation cohort (C-index 0.81), and external validation cohorts (C-index 0.70&0.66), significantly stratifying patients into distinct high- and low-risk DFS groups. Multimodal feature fusion consistently outperformed models based on single feature categories (clinical data, imaging, or radiomics alone), highlighting the synergistic value of integrating diverse data sources. By integrating multimodal data to predict DFS and recurrence risk, CerviPro provides a clinically valuable prognostic tool for LACC, offering the potential to guide personalized treatment strategies.

Cervical cancer is the most common malignancy of the female reproductive system. In 2022, an estimated 661,000 new cases were reported globally, ranking it as the fourth most prevalent cancer among women worldwide¹. In China, approximately 150,700 new cases and 55,700 deaths were recorded the same year, with both incidence and mortality rates steadily increasing over the past decades². A substantial proportion of cervical cancer cases, particularly in developing countries, are diagnosed as locally advanced cervical cancer (LACC). The standard treatment for LACC is concurrent chemoradiotherapy (CCRT), comprising external beam radiotherapy (EBRT), brachytherapy, and cisplatin-based chemotherapy. However, the disease-free survival (DFS) rate following CCRT is approximately 70%, leaving around one-third of patients at risk of treatment failure^{3–5}.

To improve survival outcomes for LACC patients, several intensified treatment strategies have been explored as adjuncts to standard CCRT. Immunotherapy^{6,7} and neoadjuvant chemotherapy^{8,9} have been shown to improve survival in LACC patients. The KEYNOTE-A18 trials reported

that integrating pembrolizumab into CCRT improved the 3-year overall survival (OS) for high-risk cervical cancer patients from 74.8 to 82.6%⁶. Despite these promising outcomes, intensified approaches result in heightened toxicity^{7,8} and substantial socioeconomic burdens. Identifying patients at high risk of treatment failure before initiating therapy could enable the delivery of targeted intensified treatments to these individuals, while sparing patients likely to respond well to standard CCRT from unnecessary escalation.

Currently, the primary method for identifying poor-prognosis patients with LACC relies on FIGO staging. Previous studies evaluating intensified treatments in addition to CCRT have also utilized FIGO staging as the enrollment criterion^{7,8}. However, significant prognostic heterogeneity has been observed among patients with the same FIGO stage^{10,11}. To overcome this limitation, we have incorporated additional prognostic risk factors for cervical cancer, such as the number and size of lymph node metastases, histology type, tumor size, squamous cell carcinoma antigen, etc^{12–16}, to develop nomogram-based prognostic

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Table 1 | Patient characteristics in the training, internal validation and external validation cohort

Characteristics	Training cohort (n = 691)	Internal Validation cohort (n = 173)	P value	External Validation cohort (n = 154)
Age, years	51 (46–59)	52 (46–60)	0.543	57(50–65)
Histological type, n (%)			0.814	
Squamous cell carcinoma	623 (90.2)	157 (90.8)		154 (100)
Non-squamous cell carcinoma	68 (9.8)	16 (9.2)		0 (0)
Follow-up duration, months	50.50 (25.73–64.57)	52.17 (25.42–63.53)	0.989	37.73 (24.72–45.02)
Max tumor diameter, cm	4.04 ± 1.67	4.05 ± 1.65	0.942	5.34 ± 1.66
FIGO stage, n (%)			0.074	
IB	85 (12.3)	17 (9.8)		5 (3.7)
IIA	41 (5.9)	12 (6.9)		7 (4.5)
IIB	297 (43.0)	76 (43.9)		40 (26.0)
IIIA	14 (2.0)	4 (2.3)		5 (3.2)
IIIB	56 (8.1)	4 (2.3)		35 (22.7)
IIIC1	155 (22.4)	49 (28.3)		49 (31.9)
IIIC2	36 (5.2)	11 (6.4)		13 (8.4)
IVA	7 (1.0)	0 (0)		0 (0)
Hemoglobin, g/L	124.83 ± 20.60	122.25 ± 21.21	0.144	120.90 ± 20.25
SCC, ng/mL	4.50 (1.50–13.30)	3.90 (1.20–13.10)	0.385	10.83 (3.70–34.33)
Pelvic lymph node metastasis, n (%)	189 (27.4)	59 (34.1)	0.079	62 (40.3)
Para-aortic lymph node metastasis, n (%)	38 (5.5)	11 (6.4)	0.662	13 (8.4)
Overall radiotherapy time, days	51 (49–56)	51 (49–57)	0.807	43 (39–55)

models^{14,16}. Compared to FIGO staging alone, these models enhance predictive accuracy. Nonetheless, clinical models are inherently restricted by the number of factors they can include and often fail to account for the heterogeneity of the tumor's intrinsic biological characteristics, limiting their overall predictive power.

Several radiomics models have been developed to predict survival outcomes in patients with LACC treated with CCRT^{17–25}. Despite their potential, these models face significant barriers to clinical implementation. First, most existing studies are constrained by small sample sizes^{17–21} and a lack of external validation^{22–24}. Only a few studies have employed large cohorts with external validation²⁵, which raises concerns about the robustness and generalizability of their findings. Second, the tumor's response to radiotherapy, commonly referred to as radiotherapy sensitivity, is a key determinant of LACC prognosis. Yet, current studies largely rely on pre-treatment imaging^{17–22,24,25}, neglecting the valuable prognostic information embedded in post-treatment tumor dynamics. Integrating imaging data acquired during or after treatment could provide critical insights into radiotherapy sensitivity, thereby improving the predictive performance of these models²³. Finally, conventional radiomics workflows often require manual delineation of regions of interest (ROIs)^{19–21,24,25}, a labor-intensive and inherently subjective process that impedes scalability and limits broader clinical adoption.

To address these challenges, this study introduces a multi-modal deep-learning-based approach that integrates pre-trained foundational models with multimodal feature fusion^{26,27}. Unlike traditional radiomics methods, our approach leverages a foundational model pre-trained through self-supervised learning on medical imaging data, offering superior feature extraction capabilities and enhanced generalizability across multi-center datasets. Furthermore, we incorporated deep features from both pre- and post-treatment CT images, clinical characteristics, and handcrafted radiomics features to develop a multimodal deep learning model for predicting recurrence risk and survival outcomes in cervical cancer patients undergoing definitive radiotherapy. The robustness and clinical applicability of this model were further validated using both internal and external cohorts, ensuring its reliability across diverse clinical settings.

Results

Patient characteristics

This study included a total of 1018 patients, distributed across the training cohort (n = 691), internal validation cohort (n = 173), and external validation cohort (n = 154). Baseline characteristics and treatment-related data are summarized in Table 1. Among all cohorts, most patients were classified as FIGO 2018 stage IIB, with squamous cell carcinoma being the most common histological subtype. The median age of the patients was 52 years. During the follow-up period (median duration: 47.3 months), 20.1% of patients experienced recurrence or disease progression. No statistically significant differences were observed in key variables such as age, FIGO stage, or tumor size between the training and internal validation cohorts ($P > 0.05$), indicating a balanced and appropriate data distribution. However, the external validation cohort showed significant differences in histological type, tumor diameter, FIGO stage, regional lymph node metastasis, and follow-up duration (Supplementary Table 1). Comprehensive details of patient demographics and clinical characteristics across different centers are summarized in Supplementary Tables 1–3. Supplementary Figs. 1 and 2 show the correlations between clinical characteristics within each cohort across different datasets, providing insights into the variations in tumor burden, treatment protocols, and follow-up durations that may influence model performance. Notably, the distribution of deep features extracted by the deep learning model, handcrafted radiomic features, and clinical characteristics was consistent across cohorts, providing a solid foundation for robust cross-center validation (Supplementary Fig. 3).

CerviPro model for prognostic prediction

In the deep neural network-based survival prediction, three categories of features were integrated: deep features, handcrafted radiomic features, and clinical characteristics. For deep features, the model extracted 8192 high-dimensional features from pre-treatment and post-treatment CT images, which were then reduced to 30 principal components through principal component analysis (PCA). This dimensionality reduction retained key prognostic information while minimizing redundancy. Handcrafted radiomic features consisted of 214 extracted parameters, encompassing shape, texture, and statistical distribution metrics. A variance-based feature

selection method was applied, narrowing the set to 147 significant features associated with survival outcomes. Clinical characteristics included nine variables: patient age, FIGO stage, tumor size, hemoglobin levels, and squamous cell carcinoma (SCC) antigen levels, among others. After standardization, all three feature categories were combined into a multimodal feature set and input into the deep learning model for survival prediction. Consistency analyses demonstrated that the distribution of these features was balanced across the training, internal validation, and external validation cohorts, establishing a strong foundation for cross-center validation. Details for the implementation of the trained network are available at: <https://github.com/yanguang-siat/CerviPro>.

Survival prediction and risk stratification

Figure 1 illustrates the Kaplan–Meier survival analysis for DFS in patients stratified into high-risk and low-risk groups based on the predictions of the CerviPro model. In the training cohort (Fig. 1A), the survival curves exhibited a clear separation ($p < 0.001$), demonstrating the model's ability to effectively distinguish between patients at varying risk levels. Similarly, in the internal validation cohort (Fig. 1B), the survival curves for the high-risk and low-risk groups remained significantly distinct ($p < 0.001$), confirming the robustness and predictive accuracy of the model within the same clinical environment. We also compared the ability of different models to distinguish between high- and low-risk groups in the training and internal validation cohorts (Supplementary Fig. 4). The results show that CerviPro outperforms the other two models in distinguishing between high- and low-risk groups. However, external validation demonstrated reduced model performance (cohort 1: $p = 0.085$, cohort 2: $p = 0.211$), highlighting important challenges in multi-center generalizability. Through systematic analysis, we identified key factors contributing to this performance gap: (1) significant differences in patient characteristics between centers, with external cohorts showing higher tumor burden (mean diameter 5.34 vs 4.04 cm, $p < 0.001$) and more advanced disease stages (66.2% vs 38.9% stage III–IV, $p < 0.001$); (2) variations in imaging protocols and quality assurance procedures; and (3) shorter follow-up duration in external cohorts (median 37.73 vs 50.82 months, $p < 0.001$).

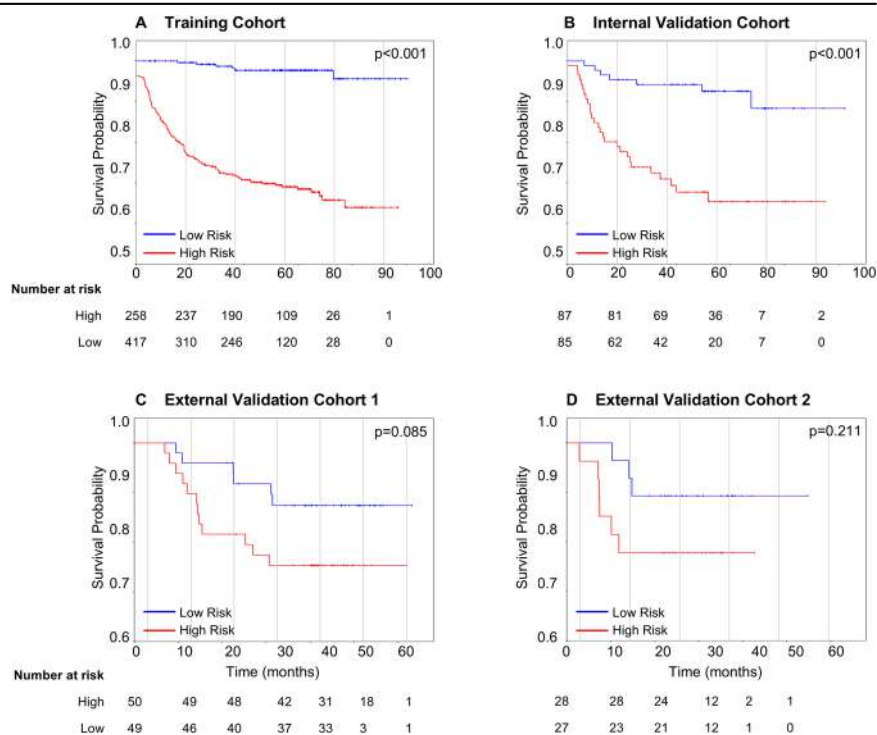
Time-dependent predictive performance

Figure 2 illustrates the performance of the CerviPro model in predicting DFS across training, internal validation, and external validation cohorts at different time points (12, 36, and 60 months). In the training cohort, the model achieved AUC values of 0.96, 0.92, and 0.87, respectively, indicating strong discriminatory ability between high- and low-risk patients, stable predictive performance, and excellent sensitivity and specificity. In the internal validation cohort, the AUC values at the same time points were 0.79, 0.80, and 0.80, slightly lower than those in the training cohort but still demonstrating high predictive accuracy, confirming the model's robustness within the same clinical center. To validate the superiority of CerviPro over other models, we compared the predictive performance of different models with the same input (Supplementary Fig. 5). The results show that at all three time points, the CerviPro model demonstrates superior predictive performance, with AUC values which are higher than those of the Cox model and the DeepSurv model. These results suggest that CerviPro consistently outperforms the other models in predictive accuracy across different time points.

However, in the external validation cohort, the AUC values dropped to 0.64–0.63, indicating a more limited predictive ability. This discrepancy may be attributed to a smaller sample size, shorter follow-up duration, differences in patient characteristics (e.g., higher tumor burden), and heterogeneity in imaging data across centers, exposing the model's limited generalizability to external datasets. Further analysis revealed a downward trend in AUC values across all cohorts as the prediction time extended to 60 months, likely reflecting the increased complexity and uncertainty associated with long-term DFS prediction. Overall, the CerviPro model demonstrated strong predictive performance in the training and internal validation cohorts but faced limitations in external validation due to differences in data distribution. These findings suggest that future studies should focus on incorporating larger multi-center datasets, improving the model's adaptability to heterogeneous data, and extending follow-up durations to enhance its clinical applicability and long-term predictive accuracy.

Fig. 1 | Kaplan–Meier survival curves for DFS prediction using the CerviPro model across training, internal validation, and external validation cohorts. Kaplan–Meier survival analysis was used to evaluate the predictive performance of the CerviPro model in stratifying patients into high-risk and low-risk groups based on DFS probabilities.

A In the training cohort, the survival curves for high-risk and low-risk groups demonstrated significant separation ($p < 0.001$). **B** Similarly, in the internal validation cohort, the model exhibited robust discrimination between the two risk groups ($p < 0.001$). **C** In external validation cohort 1, the survival curves showed a separation trend, though statistical significance was not achieved ($p = 0.085$). **D** In external validation cohort 2, a similar trend was observed, but without statistical significance ($p = 0.211$). The number of patients at risk is indicated below each plot. These results highlight the model's strong predictive performance in internal datasets and its potential applicability to external datasets, while also emphasizing the importance of addressing cohort heterogeneity and sample size in future studies.



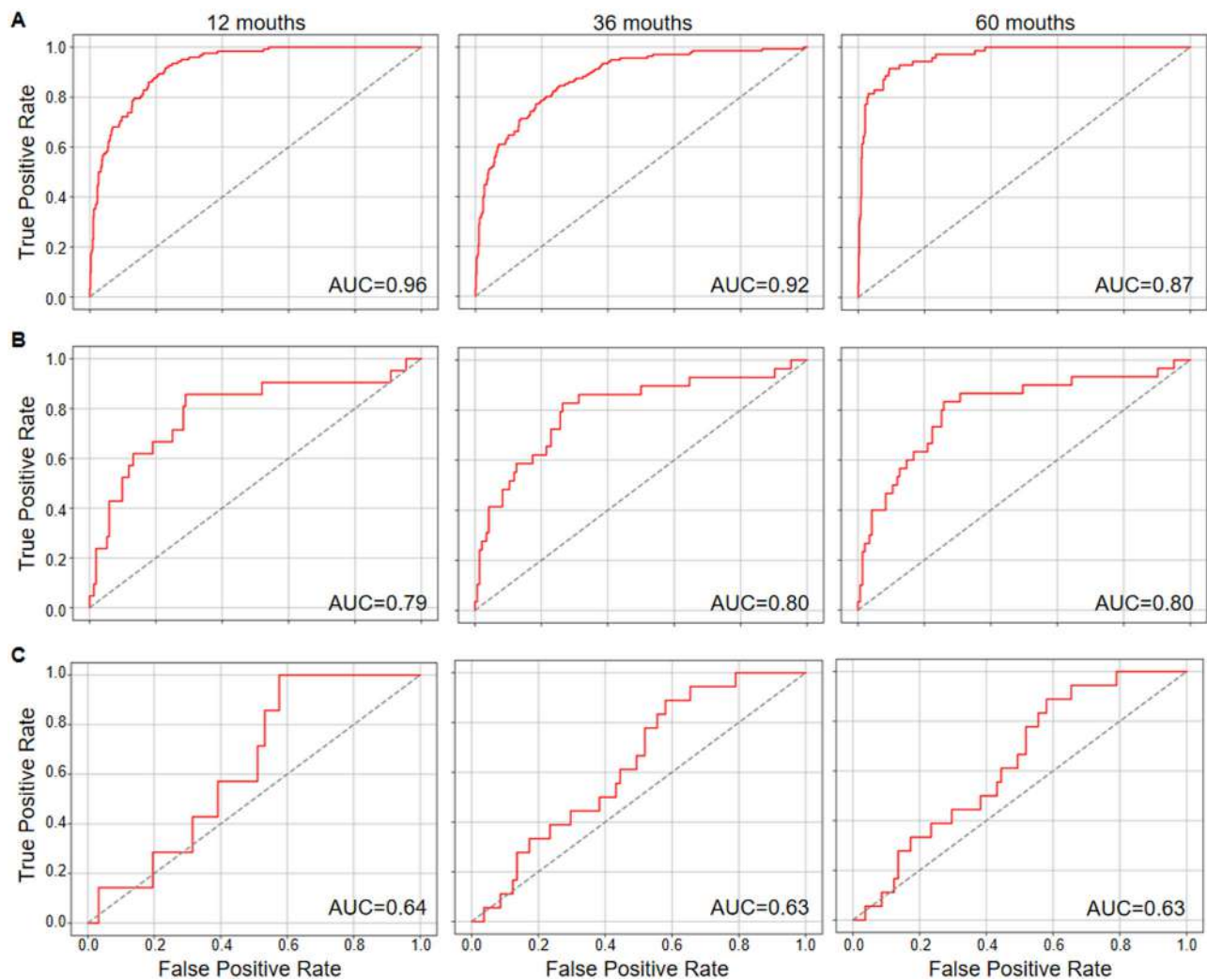


Fig. 2 | Receiver operating characteristic (ROC) curves and area under the curve (AUC) values for DFS prediction at 12, 36, and 60 months across cohorts. A In the training cohort, the CerviPro model demonstrated excellent predictive performance with AUC values of 0.96, 0.92, and 0.87 at 12, 36, and 60 months, respectively, indicating high sensitivity and specificity in stratifying patients by recurrence risk. **B** In the internal validation cohort, the model achieved AUC values of 0.79, 0.80, and 0.80 at the same time points, confirming its robustness and applicability within the

same clinical center. **C** In the external validation cohort, the AUC values were 0.64, 0.63, and 0.63, reflecting reduced performance likely due to smaller sample size, shorter follow-up duration, and differences in patient characteristics and imaging protocols across centers. These results highlight the model's strong predictive accuracy in internal datasets and suggest potential limitations in generalizability to external, multi-center cohorts, emphasizing the importance of addressing data heterogeneity and cohort variability in future research. AUC area under the curve.

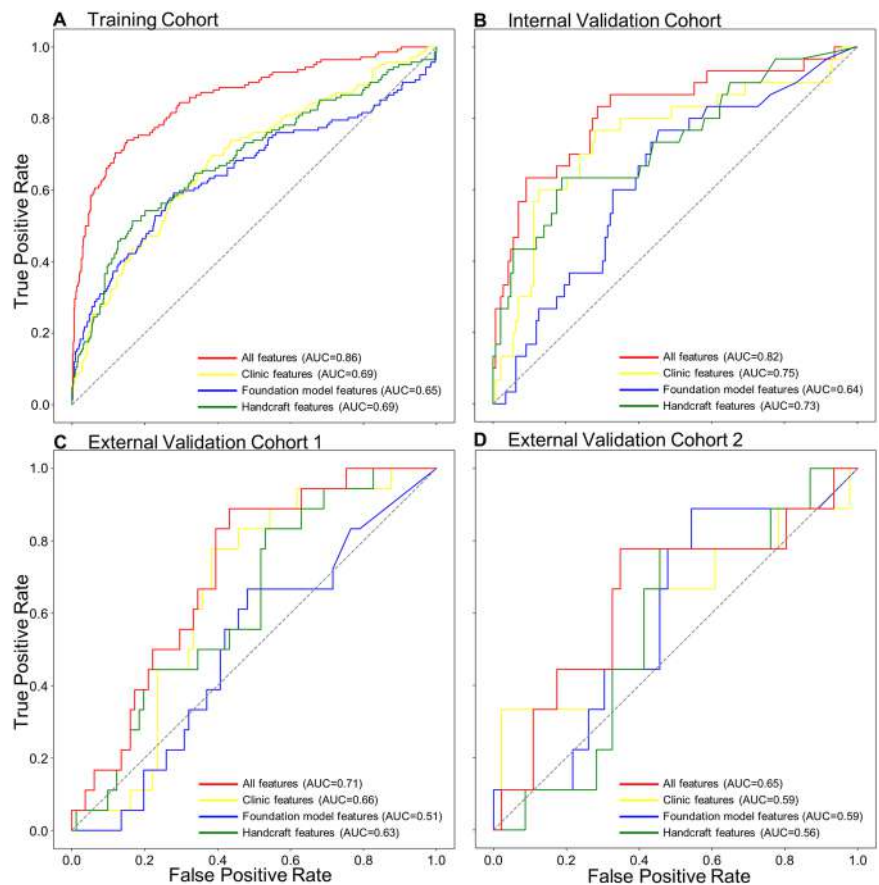
Contribution of multimodal feature fusion

Figure 3 illustrates the comparative performance of the CerviPro model using different feature combinations: clinical features, foundational model features, handcrafted radiomic features, and the integration of all three (multimodal fusion). In the training cohort (Fig. 3A), the model that incorporated all features achieved the highest AUC (0.86), significantly outperforming models based on single feature categories (clinical: 0.69, foundational: 0.65, handcrafted: 0.69). This demonstrates the synergistic effect of multimodal feature fusion in capturing complex information related to DFS. Similarly, in the internal validation cohort (Fig. 3B), the multimodal model maintained superior performance (AUC = 0.82), further validating the efficacy of the feature integration strategy. However, in external validation cohort 1 (Fig. 3C) and cohort 2 (Fig. 3D), the performance of the multimodal model declined (AUC = 0.71 and 0.65, respectively). Additionally, we conducted model updating using transfer learning techniques on a small subset of external data (20%) before validation on the remaining samples, which improved model performance in external cohorts (AUC improved to 0.76–0.79, $p < 0.001$).

Comparative analysis of prediction models

A comparative evaluation of prediction models, including Cox, DeepSurv, and CerviPro, indicated that the CerviPro model can achieve the highest concordance index (C-index) and the most significant risk stratification across all cohorts. In the internal validation cohort, the C-index of the CerviPro model was 0.806, outperforming Cox (0.663) and DeepSurv (0.717) models (Supplementary Table 4). The Kaplan–Meier analysis further confirmed that the CerviPro model achieved the most pronounced separation between high-risk and low-risk groups, demonstrating its ability to capture complex, non-linear relationships between features and survival outcomes (Fig. 1). In external validation cohorts, the performance of the Cox and DeepSurv models deteriorated significantly, likely due to overfitting and the ‘curse of dimensionality’ associated with high-dimensional feature sets. In contrast, the DeepHit-based CerviPro model demonstrated consistent, albeit reduced, performance, indicating its relative robustness to data heterogeneity and sample size constraints. At the same time, we conducted experiments to validate the predictive performance of the models using different types of feature combinations as inputs (Supplementary Tables 5–6).

Fig. 3 | Receiver operating characteristic (ROC) curves and area under the curve (AUC) values for DFS prediction using different feature sets across cohorts. A In the training cohort, the model utilizing all features (clinical, handcrafted radiomics, and foundation model features) achieved the highest AUC of 0.86, significantly outperforming models using only clinical features (AUC = 0.69), foundation model features (AUC = 0.65), or handcrafted features (AUC = 0.69). **B** In the internal validation cohort, the all-feature model also demonstrated superior performance with an AUC of 0.82, followed by clinical features (0.75), handcrafted features (0.73), and foundation model features (0.64). **C** In external validation cohort 1, the all-feature model achieved a moderate AUC of 0.71. Clinical features (0.66) and handcrafted features (0.63) provided reasonable discrimination, whereas foundation model features showed limited performance (AUC = 0.51). **D** In external validation cohort 2, the model's overall performance declined, with the all-feature model reaching an AUC of 0.65. Other feature sets performed similarly but with slightly lower AUCs: clinical features (0.59), foundation model features (0.59), and handcrafted features (0.56). These results highlight the benefit of multimodal feature fusion in internal datasets and reveal performance degradation in external cohorts, underscoring the challenges of model generalization across centers. AUC area under the curve. DFS disease-free survival.



Multi-center retraining evaluates model robustness and generalization performance

To further investigate the influence of multi-center data on model performance, we designed two retraining scenarios (A and B) and evaluated their impact separately on internal and external validation cohorts.

In Scenario A, the model was retrained using a combined dataset comprising the original training cohort and both external validation cohorts (External Cohorts 1 and 2), and then evaluated on the original internal validation cohort. The results showed minimal performance changes in Supplementary Fig. 6A, with AUCs at 12, 36, and 60 months remaining nearly identical to the original model, and a slight increase in the C-index from 0.806 to 0.808. This consistency indicates that the model maintained stable performance despite the inclusion of heterogeneous external data, highlighting its robustness within the original clinical setting.

In contrast, Scenario B focused on assessing the model's generalizability. Here, External Cohort 2 (Xijing Hospital) was added to the training set, and model performance was evaluated on the held-out External Cohort 1 (The Second Affiliated Hospital of Dalian Medical University). Notably, this retrained model achieved substantial performance gains as shown in Supplementary Fig. 6B, with the 12-month AUC increasing from 0.64 to 0.70 and the C-index rising from 0.662 to 0.700. These results suggest that the inclusion of External Cohort 2 introduced complementary information that improved the model's ability to generalize to an independent external dataset. However, despite the improvement, performance in the external cohort remained lower than in the internal set, underscoring the ongoing challenge posed by inter-center data heterogeneity.

Overall, these findings validate the model's robustness within its native data environment and demonstrate that integrating multi-center datasets can enhance generalizability. They also reinforce the necessity of developing unified, multi-institutional training frameworks to mitigate the limitations introduced by dataset variability in real-world clinical applications.

Clinical implications

These results highlight the clinical applicability of the CerviPro model for predicting DFS in cervical cancer patients undergoing definitive radiotherapy. The model's integration of pre- and post-treatment imaging data, clinical variables, and radiomic features provides a comprehensive approach to risk stratification, enabling personalized treatment strategies. However, the reduced performance in external validation cohorts underscores the need for further refinement. Future research should focus on addressing data heterogeneity through standardized imaging protocols, incorporating larger multi-center datasets, and extending follow-up durations to enhance the model's generalizability and long-term predictive accuracy. Additionally, the inclusion of real-world data and prospective validation studies will be critical for translating the CerviPro model into routine clinical practice.

Discussion

In this study, we developed and validated a deep learning-based prognostic model (CerviPro) that integrates longitudinal CT imaging, handcrafted radiomic features, and clinical variables to predict DFS in patients with LACC receiving definitive radiotherapy. Our findings highlight CerviPro's potential to surpass conventional models (such as Cox proportional hazards and DeepSurv) in internal validation, likely owing to its capacity for comprehensive feature extraction and fusion. By including post-treatment imaging, CerviPro captures dynamic tumor responses, an aspect often overlooked in traditional radiomics models, thereby refining risk stratification and providing a basis for more individualized treatment strategies.

In recent years, there have been significant advancements in the treatment of LACC. Researchers have explored various intensified treatment strategies aimed at enhancing survival outcomes for patients. Notably, immunotherapy (KEYNOTE-A18 trial)⁶⁷ has demonstrated its ability to improve survival in individuals with LACC. OUTBACK trial revealed that adjuvant chemotherapy after CCRT increased toxicities without enhancing

survival outcomes²⁸. While these studies made efforts to include high-risk patients, their screening methods predominantly depended on staging, which often fell short in identifying true high-risk patients. For instance, the KEYNOTE-A18 trial targeted patients diagnosed with FIGO 2014 stage IB2–IIB with node-positive disease or stage III–IVA regardless of nodal status⁶⁷. Similarly, OUTBACK trials incorporated subjects with FIGO 2008 stage IB1 disease accompanied by nodal involvement, as well as stages IB2, II, IIIB, or IVA disease²⁸. Despite this approach, staging alone offers limited predictive value and lacks the precision needed to effectively identify high-risk individuals^{10,11}. In this study, our newly developed CerviPro model stands out as a robust tool for stratifying patients into high-risk and low-risk groups, with significantly different DFS. During internal validation, the CerviPro model achieved an AUC of 0.79–0.80 and a C-index of 0.81, indicating its reliability and accuracy. By using the CerviPro model for risk stratification, it is possible to more precisely target high-risk patients who are likely to experience treatment failure, and provide them with intensified treatment options, such as immunotherapy.

Despite strong performance in the training and internal validation cohorts, CerviPro showed reduced predictive accuracy in the external validation cohorts. Multiple factors may explain this discrepancy, including heterogeneity in imaging protocols, variability in patient characteristics (e.g., more advanced tumor stages and greater tumor burden), and relatively short follow-up in external centers. Such variability emphasizes the critical need for standardized imaging acquisition, careful protocol harmonization, and larger multicenter datasets to improve generalizability. Incorporating additional modalities (e.g., magnetic resonance imaging or genomic data), as well as applying transfer learning or model recalibration to external datasets, could further enhance robustness.

Several features of CerviPro underscore its clinical relevance. First, it relies on automated segmentation methods and standard CT imaging, facilitating streamlined integration into routine clinical workflows. Second, by focusing on commonly collected clinical data (e.g., hemoglobin, SCC antigen) and combining it with radiomic and deep imaging features, CerviPro remains broadly applicable across diverse clinical settings. Finally, its

ability to discriminate high- versus low-risk groups offers a practical framework for tailoring treatment intensity—potentially guiding decisions on adjuvant therapy, thereby sparing low-risk patients unnecessary intensified treatment.

However, important limitations remain. We did not systematically explore how specific CT parameters (e.g., slice thickness, reconstruction algorithm) influenced model performance, and we did not include molecular or pathological biomarkers that could further refine prognostication. The black-box nature of deep learning also raises questions about interpretability, underscoring the need for user-friendly interfaces and explainable AI approaches to foster clinical trust and adoption. Additionally, shorter follow-up times in external centers limited our ability to assess long-term outcomes. Future work should address these constraints through prospective, multicenter trials with extended follow-up, standardized imaging protocols, and incorporation of domain-specific imaging or molecular features.

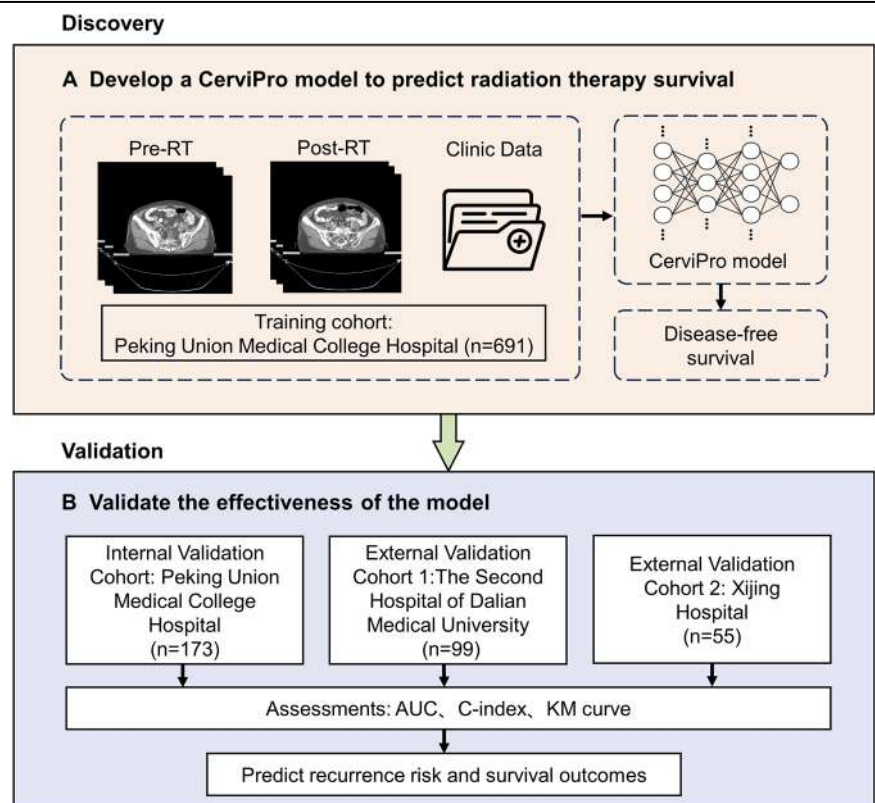
In conclusion, CerviPro demonstrates strong potential to improve DFS prediction and guide personalized treatment in LACC. By integrating longitudinal imaging data, radiomic features, and clinical parameters, our model offers an enhanced approach to risk stratification. Larger prospective trials, harmonized imaging procedures, and inclusion of complementary biomarkers could further validate and refine this framework, ultimately paving the way for more precise care in cervical cancer management.

Methods

Study design and patients

Figure 4 illustrates the overall study design and validation workflow, encompassing the development and multi-center validation of the CerviPro model. During the model development phase, the training cohort (n = 691) from Peking Union Medical College Hospital (Cohort 1) was utilized to integrate pre- and post-treatment CT imaging data with clinical characteristics. A multimodal feature extraction process, combining deep learning-based automatic segmentation, handcrafted radiomic features, and key clinical variables, was employed to construct the CerviPro model for

Fig. 4 | Study design and workflow for CerviPro model development and validation. The figure outlines the development and multi-center validation of the CerviPro model for predicting DFS in cervical cancer patients undergoing definitive radiotherapy. **A** Model development: Using the training cohort (n = 691, Peking Union Medical College Hospital), the model integrated pre- and post-treatment CT imaging, clinical data, and radiomic features through multimodal feature fusion. **B** Model validation: The model was validated using an internal cohort (n = 173) and two external cohorts (n = 99 and n = 55) from independent centers. Validation metrics, including AUC, C-index, and Kaplan–Meier analysis, assessed the model’s performance, robustness, and generalizability. RT Radiotherapy. AUC area under the curve. C-index concordance index. KM Kaplan–Meier.



predicting DFS in cervical cancer patients undergoing definitive radiotherapy. In the validation phase, the model's performance was assessed using multi-center data, including an internal validation cohort ($n = 173$, Peking Union Medical College Hospital) and two external validation cohorts: Xijing Hospital (Cohort 2, $n = 55$) and The Second Affiliated Hospital of Dalian Medical University (Cohort 3, $n = 99$). Detailed patient selection criteria are provided in Supplementary Fig. 7. Metrics such as AUC, C-index, and Kaplan–Meier survival analysis were used to evaluate the model's accuracy, robustness, and generalizability. The figure clearly depicts the complete workflow from feature extraction to model validation, highlighting the systematic study design and the rigor of multi-center validation.

This study was approved by the ethics committees of Peking Union Medical College Hospital, Xijing Hospital, and the Second Affiliated Hospital of Dalian Medical University in compliance with the ethical principles outlined in the Declaration of Helsinki. The requirement for informed consent was waived by all three ethics committees based on the following considerations: (1) the study posed no more than minimal risk to participants; (2) waiver of consent would not adversely affect subjects' rights and welfare; (3) the research involved no commercial interests; (4) participants could no longer be contacted. All patients had previously provided general consent for the potential research use of their medical records and imaging data.

Treatment and follow-up

In this study, all patients received standard definitive radiotherapy regimens, which included a combination of EBRT and brachytherapy. EBRT was delivered using fixed-field intensity-modulated radiotherapy (IMRT), volumetric modulated arc therapy (VMAT), or helical tomotherapy. The clinical target volume (CTV) encompassed the primary tumor, cervix, uterus, upper vagina, parametrium, and pelvic lymph node regions. For cases demonstrating para-aortic or common iliac lymph node involvement, the para-aortic lymph node region was incorporated into the CTV. In patients presenting with multiple positive pelvic lymph nodes, prophylactic para-aortic lymph node irradiation was selectively administered. The proportion of patients receiving para-aortic lymph node irradiation across the three participating centers is detailed in Supplementary Table 7. Treatment plans were developed based on CT simulation scans, with a prescribed total dose of 50.4 Gy to the planning target volume (PTV) administered in 28 fractions. To minimize the impact of tumor shrinkage on EBRT positioning, patients at Peking Union Medical College Hospital received second CT simulation and radiotherapy planning after 20 fractions of EBRT. And the second CT simulation images were used as post-treatment CT imaging in the radiomics model.

Brachytherapy was delivered with CT/MRI-guided adaptive brachytherapy, or orthogonal X-rays-based 2D brachytherapy, with 30–36 Gy in 5–6 fractions delivered to high-risk clinical target volume (HRCTV) or point A.

The first-line regimen of concurrent chemotherapy was cisplatin weekly.

Upon completing radiotherapy, all patients were enrolled in a structured follow-up program, which comprised medical history reviews, physical examinations, laboratory tests, and imaging evaluations. The follow-up schedule included assessments every three months during the first two years post-treatment, every six months for the subsequent three years, and annually thereafter. Follow-up data were meticulously recorded and updated in electronic medical records at each participating center. The detailed treatment and follow-up approach was described in previous articles^{3,4}.

Across the three participating centers, the follow-up process for patients completed at different times: Peking Union Medical College Hospital completed its follow-up by April 2020, The Second Affiliated Hospital of Dalian Medical University by September 1, 2023, and Xijing Hospital by April 29, 2024. Throughout the follow-up period, every event associated

with disease recurrence, progression, or mortality was comprehensively documented.

Imaging acquisition and tumor segmentation

In this study, pelvic CT images for all patients were acquired using high-performance multi-slice spiral CT scanners at the three participating centers. CT images were acquired using 16-channel CT scanner (Philips Brilliance Big Bore CT, Philips, Bothell, WA, USA) in Center 1, and 64-slice CT scanner (Siemens SOMATOM Definition AS Large Bore, Siemens Healthcare, Germany) in center 2 and Center 3. The imaging data were standardized and stored in DICOM format. Imaging data from different centers underwent gray-scale normalization, spatial resolution matching, and contrast enhancement correction to minimize inter-center variability in scanning parameters.

In our study, the clinical target volume (CTV)—encompassing the primary tumor, cervix, uterus, parametrial tissues, upper vagina, and pelvic lymphatic drainage regions—served as the region of interest (ROI). A deep learning-based automatic segmentation network was employed to delineate the CTV. The model was trained using images from 423 patients from Peking Union Medical College Hospital. The flow chart of patient inclusion was shown in Supplementary Fig. 8. The CTVs employed for training, which correspond to the ROI, were the ones used clinically during patient radiotherapy. Initially, these CTVs were delineated by 9 board-certified radiation oncologists, including 6 attending physicians and 3 residents, and were subsequently reviewed and confirmed by 5 chief or deputy chief physicians. We used a 3D U-Net architecture (encoder–decoder with skip connections) trained with a composite Dice and cross-entropy loss. Input CT images were clipped to $[-200, 300]$ Hounsfield units, Z-score normalized, resampled to $1.5 \times 1.5 \times 3 \text{ mm}^3$, and rigidly registered to a pelvic atlas. Training was performed on overlapping $160 \times 160 \times 96$ patches using stochastic gradient descent (momentum 0.99, weight decay 0.001). During inference, patch predictions were stitched together, and small components (<10 voxels) were removed. This approach achieved a mean Dice score of 0.87 ± 0.04 on the test data for segmentation of pelvic lymphatic regions.

Model development

Figure 5 systematically illustrates the development process of the CerviPro model, detailing the implementation of multimodal feature fusion and its application in survival prediction. The CerviPro model was developed using three key components: (1) automated segmentation from pre- and post-treatment CT images. (2) extraction of comprehensive features including deep learning-based imaging features, radiomics features, and clinical variables. For deep learning features, we utilized a pre-trained self-supervised foundation model to extract 8192 high-dimensional features, which were then reduced to 30 principal components using principal component analysis (PCA). The number of components was determined based on preliminary experiments evaluating different dimensionalities, with 30 components selected as they explained 76.78% of the total variance (Supplementary Fig. 9). For handcrafted radiomics features, a total of 214 features—including shape, texture, and first-order statistics—were extracted within the CTV region. After variance-based feature selection, 147 features were retained. For clinical variables, nine factors were included to complement imaging-derived features. These encompassed demographic characteristics (age), disease-related parameters (FIGO stage, tumor size), and biomarkers (hemoglobin level, SCC antigen level). All variables were normalized to ensure compatibility with other feature types. And (3) we developed a deep learning-based fusion model for survival prediction, which integrates deep learning features, handcrafted radiomic features, and clinical variables into a unified framework. The fusion process is implemented using a multi-layer perceptron (MLP) designed to capture complex nonlinear relationships among multimodal inputs. The model was trained on a cohort of 691 patients and validated on both an internal cohort ($n = 173$) and an external cohort ($n = 154$).

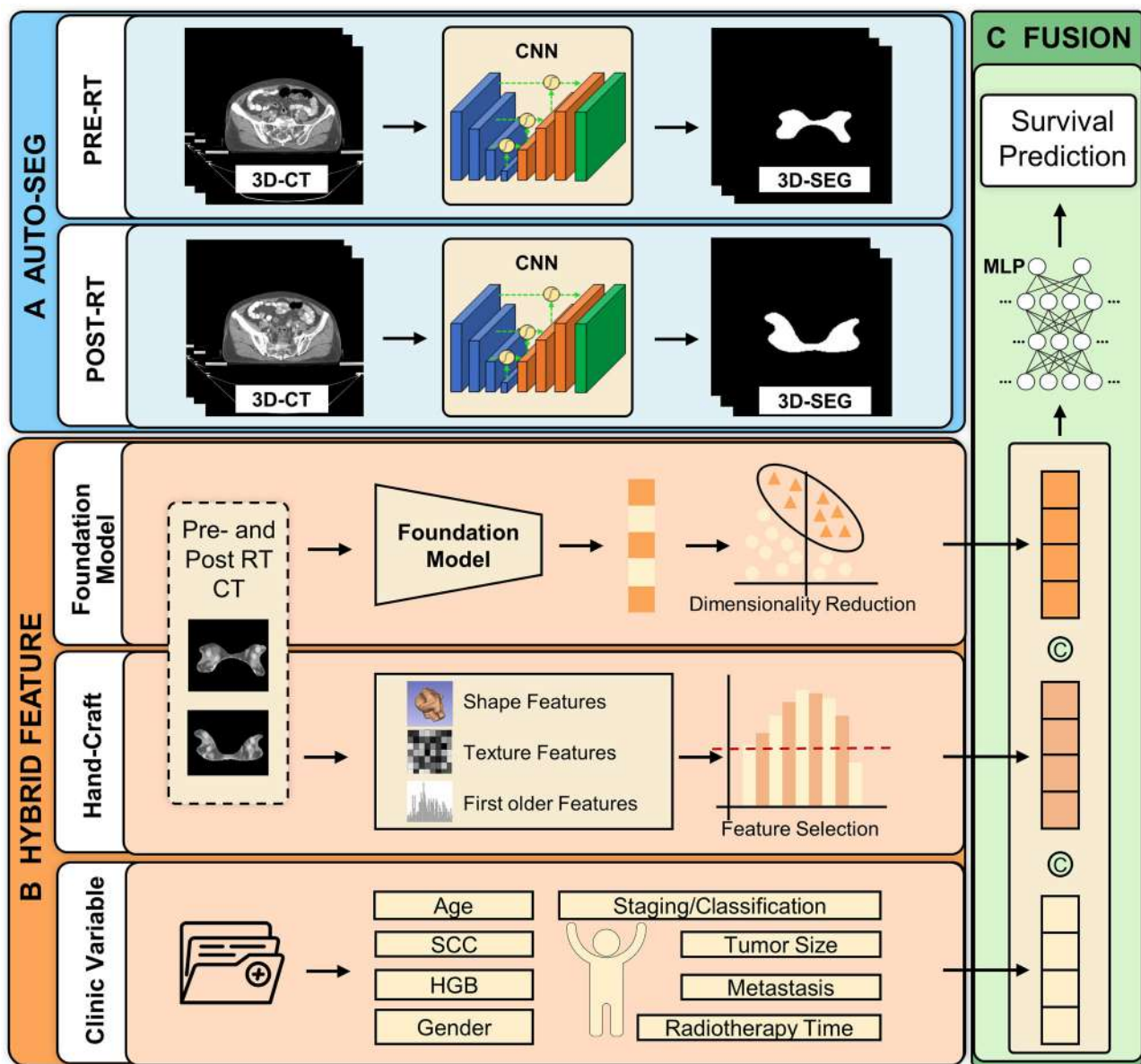


Fig. 5 | Development framework of the CerviPro model for predicting DFS in cervical cancer patients undergoing definitive radiotherapy. The figure illustrates the workflow of the CerviPro model, which integrates multimodal features for survival prediction. **A** Automatic segmentation (AUTO-SEG) module: Pre- and post-radiotherapy (PRE-RT and POST-RT) 3D-CT images were processed using CNNs to automatically segment the tumor and related anatomical structures into 3D regions (3D-SEG), providing high-quality input for feature extraction. **B** Hybrid feature extraction module: Three categories of features were extracted and integrated: (1) deep features, derived from a pre-trained foundational model applied to

pre- and post-treatment CT images, reduced to 30 principal components via principal component analysis; (2) handcrafted radiomic features, including 214 shape, texture, and first-order statistical features, narrowed to 147 significant features via variance-based selection; and (3) clinical variables, comprising 9 key factors such as age, FIGO stage, tumor size, hemoglobin levels, and SCC levels. **C** Feature fusion module: A multimodal feature set, combining deep features, handcrafted radiomic features, and clinical variables, was input into a MLP for survival prediction. RT Radiotherapy. CNN convolutional neural network. MLP multi-layer perceptron. SCC squamous cell carcinoma antigen. HGB hemoglobin.

Evaluation of the prognostic prediction model for radiotherapy outcomes

We systematically validated the prognostic accuracy of the deep learning model in predicting DFS for cervical cancer patients undergoing definitive radiotherapy by assessing its discrimination performance. Discrimination was quantified using Harrell's concordance index (C-index), which measures the accuracy of the model in ranking individual patient outcomes. To further assess the discrimination ability of the model, we stratified patients in the training cohort into high-risk and low-risk groups using the median predicted survival score as a cutoff. The same cutoff value was applied to the internal and external validation cohorts to ensure the robustness of the model across independent

datasets. Additionally, we compared the predictive performance of the deep learning-based multimodal feature fusion model with traditional survival analysis models, such as the Cox proportional hazards model and the DeepSurv model. These comparisons demonstrated the superior ability of the deep learning model to capture complex non-linear relationships and integrate multimodal information.

Statistical analysis

Continuous variables were assessed for normality using the Shapiro-Wilk test. Variables following a normal distribution were presented with mean $\pm$ standard deviation (SD) and compared between groups using independent sample *t* tests, while non-normally distributed variables were

presented with median (interquartile range, IQR) and analyzed using the Mann–Whitney U test. For categorical variables, group differences in proportions were evaluated using the Chi-square test or Fisher’s exact test, depending on the data characteristics. To assess the discriminatory performance of the predictive model, receiver operating characteristic (ROC) curve analysis and Harrell’s concordance index (C-index) were employed. Statistical significance of differences in area under the curve (AUC) values between models was tested using the DeLong test. All statistical tests were two-sided, and a *p* value of less than 0.05 was considered statistically significant. Statistical analyses were performed using R software (version 4.2.0) and SPSS software (version 27.0).

Data availability

The data that supports the findings of this study are available from the corresponding authors with a signed data access agreement. The raw image data are not publicly available because they contain sensitive information that could compromise patient privacy.

Code availability

The code is available online in: <https://github.com/yangguang-siat/CerviPro>.

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Author contributions

X.H. and X.L. had full access to all the data in the study and are responsible for the integrity of the data and the accuracy of the data analysis. K.H. conceived and designed the study. W.W., Y.L., L.W., and X.X. acquired the data. GY implemented quality control of the data and the algorithms, and developed, trained and applied the artificial neural network. C.Z. and Z.P. performed the statistical analyses. G.Y. drafted the manuscript. WW revised the manuscript. All authors analyzed and interpreted the data, contributed to manuscript preparation, read and approved the final manuscript.

Competing interests

The authors declare no competing interests.

Additional information

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